

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0906

Washburn 138 kV
July 25, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0906. The Interconnection Customer proposes a 69.4 MW solar photovoltaic facility interconnecting at the Washburn 138 kV substation in the Pennsylvania area. The request was received on December 16, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 317 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,023,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Hemet – Woodville 138.0 kV, the Hulbert – Southside Generating Station 138.0 kV loads to 101.1% of its 270 MVA rating. Project DFAX: 10.8%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hulbert – Southside Generating Station 138.0 kV	Reconductor line	\$2,099,000	49% (\$1,023,000)
	Total allocated to project		\$1,023,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.